

Reader: You mentioned that the existence of the limit of a function $\left(\frac{f(x)}{g(x)}\right)$ at a , when both $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$ (the existence of the limit of the type $\frac{0}{0}$), is very important for further considerations. Why?

Author: The point is that one of the most important concepts in calculus, namely, that of derivative, is based on the limit of the type $\frac{0}{0}$. This will be clear in the subsequent dialogues.

Dialogue Eight: Velocity

Author: We are practically ready to tackle the concept of a derivative. This concept, alongside with the concepts of the limit of numerical sequence and the limit of function, is one of the most important special concepts in calculus.

We may approach the concept of a derivative by considering, for instance, a quantity widely used in physics: the instantaneous velocity of nonuniform motion of a body.

Reader: We have been familiarized with this notion when studying kinematics in the course of physics, or, to be precise, the kinematics of nonuniform motion in a straight line.

Author: Exactly. What is your idea of the instantaneous velocity?

Reader: The instantaneous velocity of a body is defined as the velocity of a body at a given moment of time (at a given point of its trajectory).

Author: And what is your idea of the velocity at a given moment of time?

Reader: Well, I see it as If a body moves uniformly, at different moments of time its velocity remains the same. If a body moves nonuniformly (accelerating or decelerating), its velocity will, in the general case, vary from moment to moment.